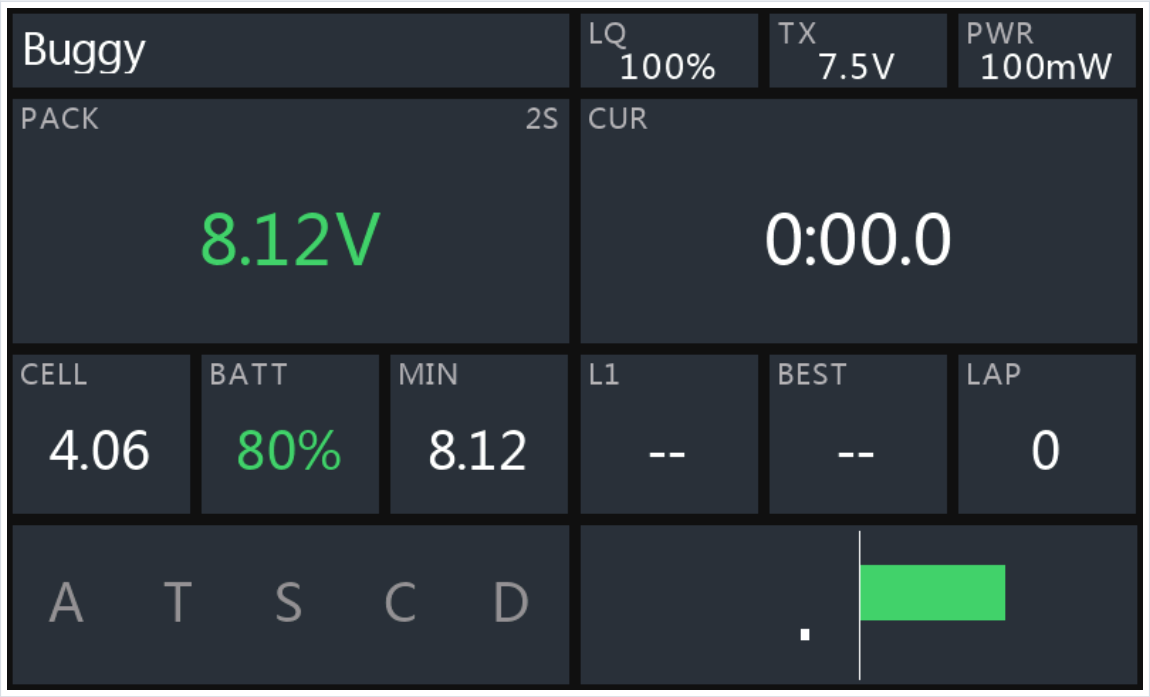
Trackside

Race dashboard and drive aids
for the FrSky Velox VX10



The Trackside dashboard on a VX10

Setup guide | Version 0.2.0 | 31 October 2026

Before you start

This guide sets up Trackside on a car model, from a fresh model through to the dashboard. It takes about ten minutes. Every screenshot is from the VX10.

You need	
Radio	FrSky Velox VX10 with Ethos 26.2 or later
Software	Trackside 0.2.0 (the **trackside** folder)
Model	A car model. Section 2 creates one with the Ethos wizard.

How Trackside connects to your car

Trackside adds these items to your radio:

Part	What it is	What it does
Throttle Aid and **Steer Aid**	Lua sources	Each outputs a multiplier for one channel. Your mixes keep the normal stick line and add one **Multiply** line with the aid after it.
Rock (optional)	Lua source	Rocks a car off its roof: while you hold the Rock Switch it pulses the throttle forward and reverse. Uses one **Add** line on the throttle channel (section 4e).
Trackside	Telemetry sensor	Created automatically. Reads 1 while Trackside runs. A logic switch built on it turns the aid lines off if Trackside stops or is missing.
Trackside	System tool	In the System menu. Holds all Trackside settings and **Check setup**, which proves the aid mixes work.
Trackside	Widget	The race dashboard.

Because the stick line stays in place, the car is always driven by your trigger and wheel. The aids can only reduce them. If the Trackside scripts are missing, the safety switch from section 4c turns the aid lines off and the car drives on the plain stick. If Trackside hits an error, the aids let the stick straight through.

> **The aids start OFF.** Every aid is OFF in a new model, so the car drives exactly as before until you turn one on (section 7). Turn them on one at a time and try each on the bench first.

> **Safety.** Make mixer changes on the bench with the car's motor disconnected or the radio's RF output off. Check failsafe still works afterwards.

1. Install Trackside

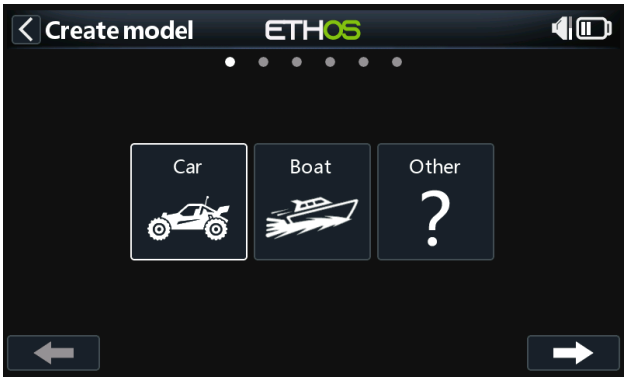
1. Connect the VX10 to your computer by USB and choose **USB storage** on the radio.
2. Copy the **trackside** folder into the radio's **SCRIPTS** folder, so you have **SCRIPTS/trackside/main.lua**.
3. Eject the radio drive safely and unplug it. The radio loads Trackside on its next start.

> Your settings are saved in **SCRIPTS/trackside.user**, a separate folder. Updating Trackside replaces only **SCRIPTS/trackside**, so your settings are kept.

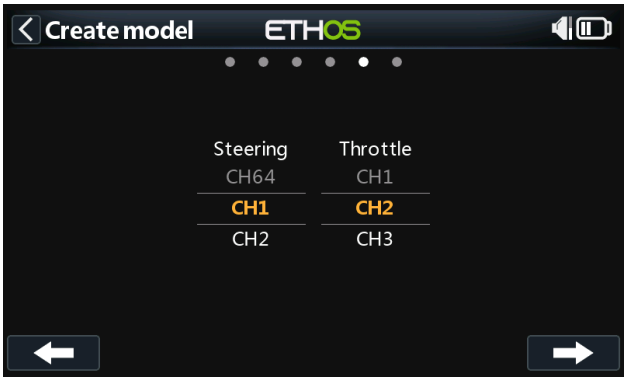
2. Create a car model

Skip this if you already have a car model with Steering on CH1 and Throttle on CH2. Otherwise create one with the Ethos wizard: **Model > Model select > +**.

1. Choose **Car**.
2. Step through the engine, steering and brake pages. The defaults are fine for most cars.
3. On the channel page, keep **Steering = CH1** and **Throttle = CH2**. The rest of this guide uses these.
4. Give the model a name and finish the wizard.



Choose Car



Steering CH1, Throttle CH2

3. Turn on Throttle Aid and Steer Aid

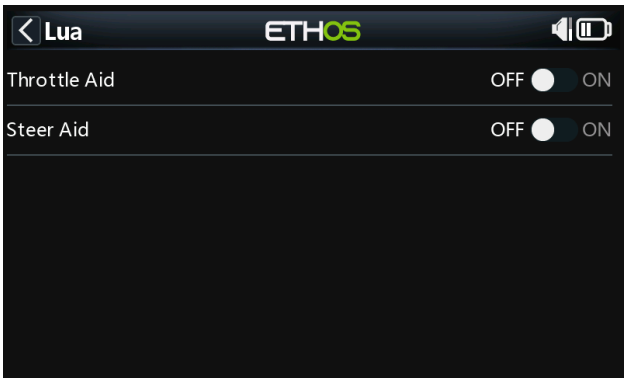
Ethos only runs a Lua source, and only lists it when you pick a mix source, after you turn it on for the model. Do this once per model.

1. Open **Model** and swipe to the second page.
2. Tap **Lua**.
3. Turn **Throttle Aid** and **Steer Aid** ON. Trackside then also creates its **Trackside** telemetry sensor, used in section 4c.

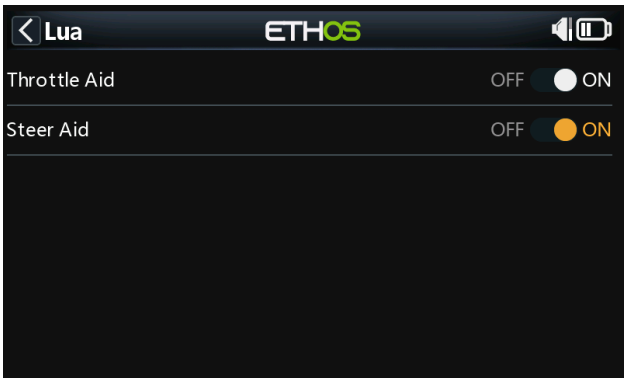
The **Rock** source is listed here too. Leave it OFF unless you want Rock (section 4e).



Model menu, page 2: the Lua page



Both start OFF

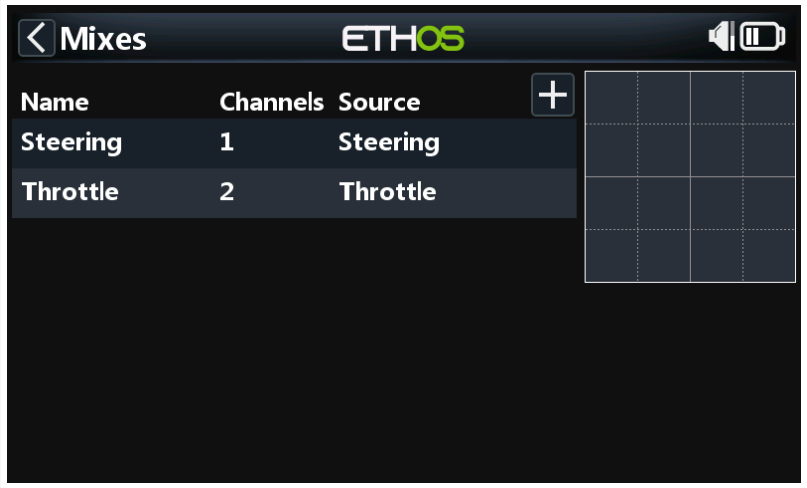


Turn both ON

4. Add the aid mixes

The car wizard made two mixes: **Steering** (CH1) and **Throttle** (CH2). **Leave them exactly as they are.** You add one more mix after each.

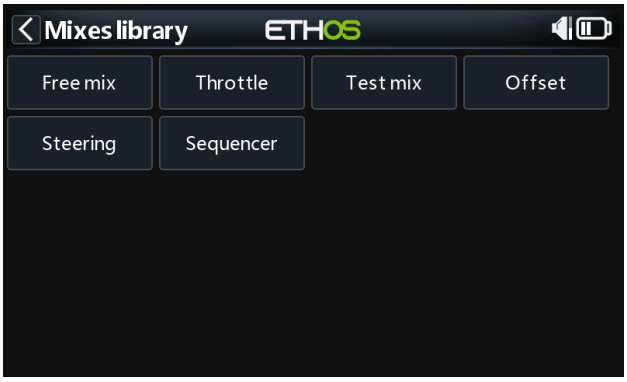
> **Don't** change the source of the existing Steering or Throttle mix to an aid. The channel would then follow the aid's fixed value instead of your stick, and nothing would move.



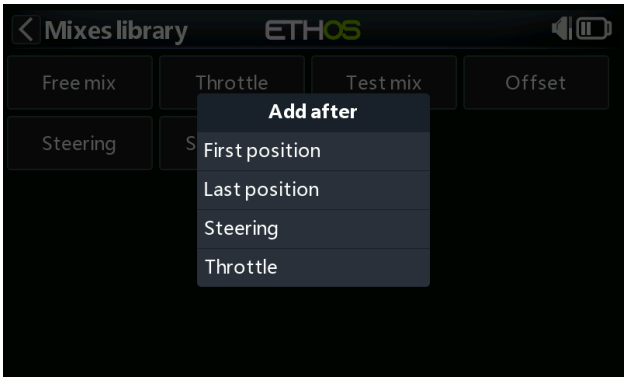
Model > Mixes: the two mixes from the wizard

4a. Throttle

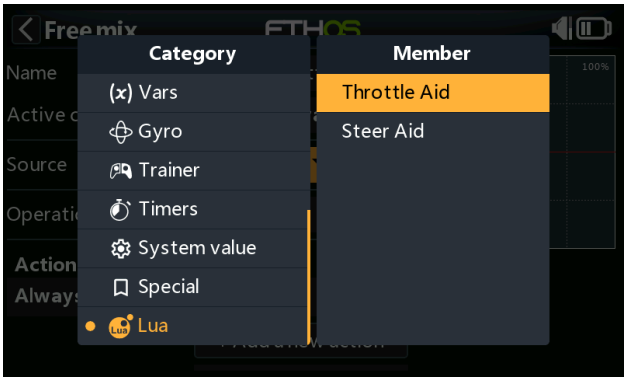
1. In **Mixes**, tap **+**, choose **Free mix**, then **Add after: Throttle**.
2. Optional: set **Name** to *Throttle Aid* so it's easy to spot in the list.
3. Tap **Source**, scroll to the **Lua** category and pick **Throttle Aid**.
4. Set **Operation** to **Multiply**.
5. Scroll down and set **Output** to **CH2 (Throttle)**. A new Free mix starts with Output = **---** and does nothing until you set it.



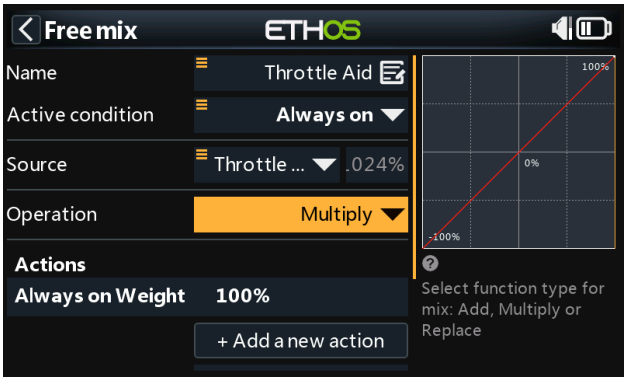
+ opens the mix library: choose Free mix



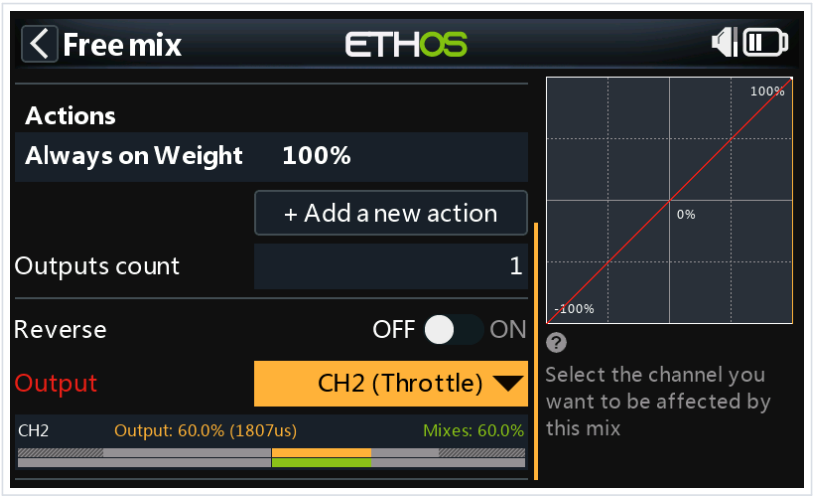
Add after: Throttle



Source: Lua > Throttle Aid



Operation: Multiply

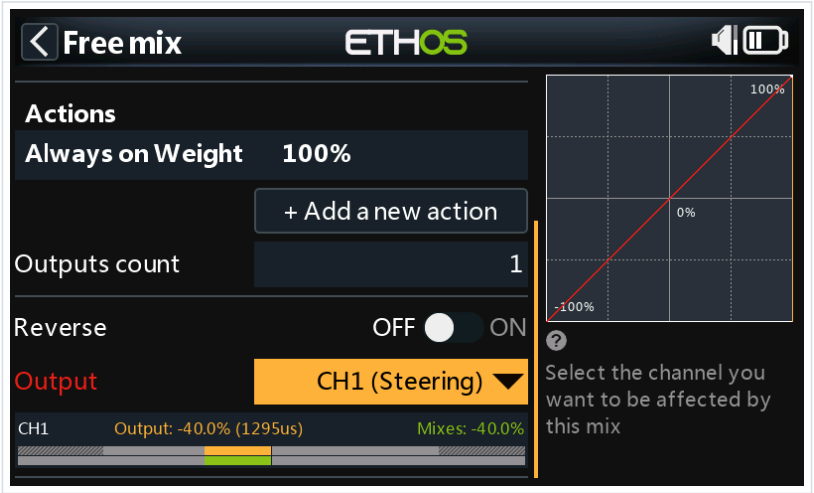


Output: CH2. The bar shows the result: throttle 60% gives CH2 60%.

> The grey box next to Source shows about **1024%**. That's Ethos showing the aid on its internal scale, and it's expected. What counts is the channel output.

4b. Steering

Repeat the same steps with these changes: **Add after: Steering**, source **Lua > Steer Aid**, Operation **Multiply**, Output **CH1 (Steering)**.



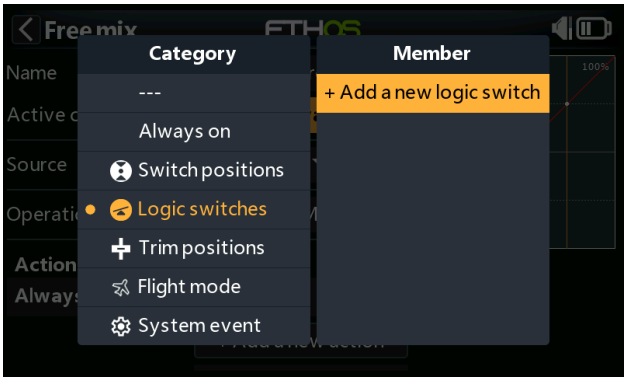
Steer Aid on CH1: wheel at -40% gives CH1 -40%

4c. Add the safety switch

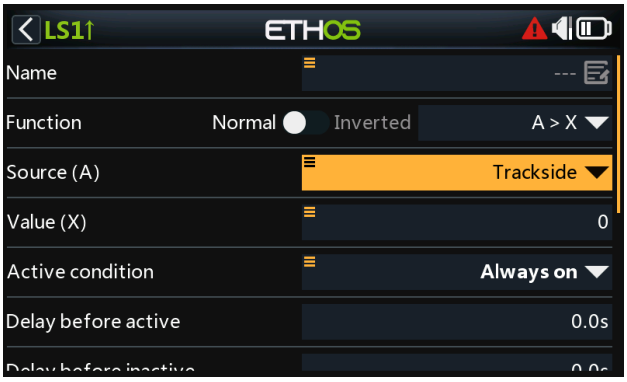
This makes the two aid lines run only while Trackside is loaded. You create one logic switch and use it as the **Active condition** of both aid lines.

> **Don't skip this.** If the Trackside scripts are missing (not copied, a new SD card, a failed update) or stops, Ethos treats each aid as zero or holds its last value. An aid line without the safety switch then multiplies its channel by zero: **no steering and no throttle**. With the switch, the aid lines turn off and the car drives on the plain stick.

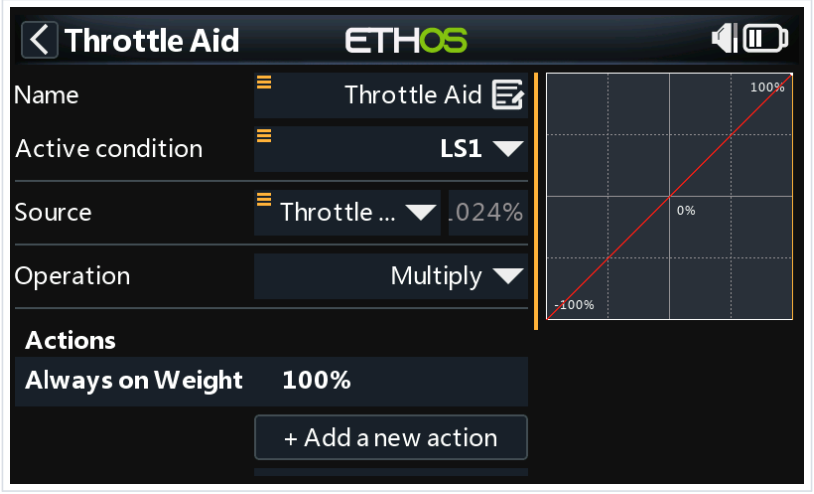
1. Open the **Throttle Aid** mix: tap it, then **Edit**.
2. Tap **Active condition**, choose **Logic switches**, then **+ Add a new logic switch**.
3. Set **Function** to **A > X** and **Source (A)** to **Telemetry > Trackside**. Leave **Value (X)** at 0. The title **LS1** turns green while Trackside is running.
4. Tap **<** to go back. The mix's Active condition now reads **LS1**.
5. Open the **Steer Aid** mix and set its **Active condition** to **Logic switches > LS1**.



Active condition: Logic switches > + Add a new logic switch



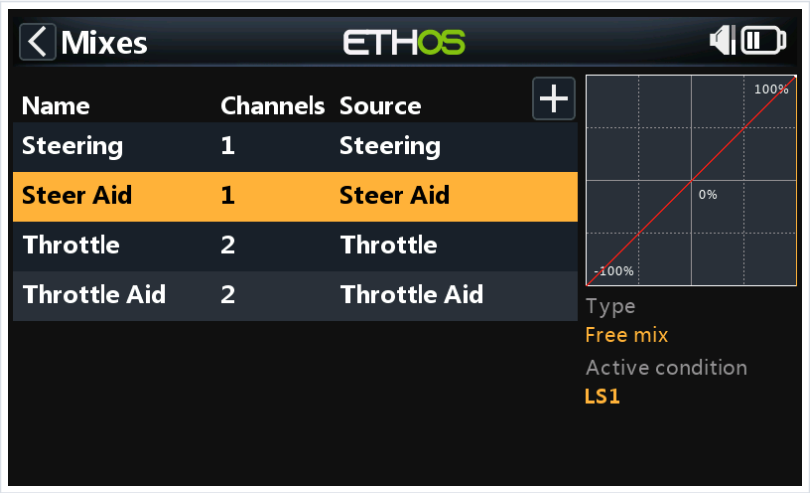
LS1: A > X, Source (A) Trackside sensor, Value (X) 0



The aid line with Active condition LS1

4d. Check the order

Each stick mix must be followed by its aid. An aid line placed **above** its stick mix does nothing. Use exactly one aid line per channel: a second Multiply line with the same aid applies it twice.



Finished: Steering then Steer Aid (CH1), Throttle then Throttle Aid (CH2), both aids on LS1

Channel	Mix	Source	Operation	Active condition
CH1	Steering	Analogs > Steering (the wheel)	Add	Always on
CH1	Steer Aid	Lua > Steer Aid	Multiply	LS1
CH2	Throttle	Analogs > Throttle (the trigger)	Add	Always on
CH2	Throttle Aid	Lua > Throttle Aid	Multiply	LS1
CH2	Rock (optional)	Lua > Rock	Add	Always on

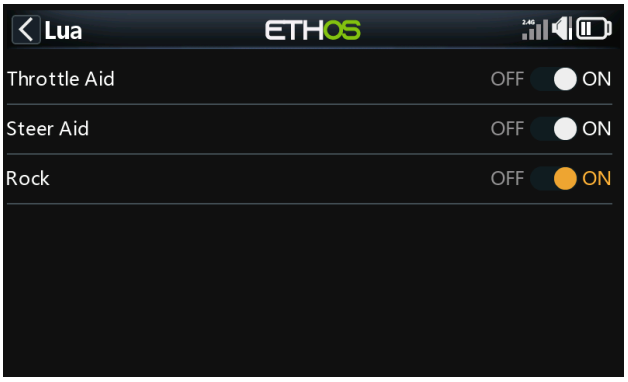
To reverse a channel, use **Model > Channels**. The wizard's stick mixes have no reverse option, and a reverse on the aid line flips the whole channel.

4e. Rock line (optional)

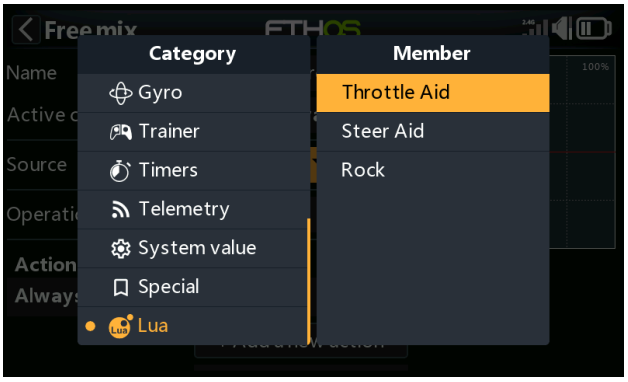
Rock gets a car off its roof. While you hold the Rock Switch, Trackside pulses the throttle forward and reverse. The pulses get longer and stronger with each rock, so one of them matches the way your car rocks. Skip this section if you don't want it.

> **Reverse must be on in the ESC.** Rock can't make an ESC with reverse disabled go backwards: the car will only brake. Try Rock on the bench first, with the car on a stand.

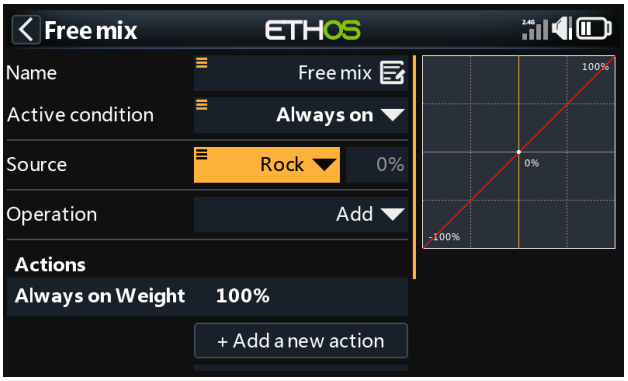
1. In **Model > Lua**, turn **Rock** ON.
2. In **Mixes**, tap **+**, choose **Free mix**, then **Add after: Throttle Aid** (or **Last position**). Optional: name it *Rock*.
3. Tap **Source**, scroll to the **Lua** category and pick **Rock**.
4. Leave **Operation** on **Add** and the weight at 100%.
5. Scroll down and set **Output** to **CH2 (Throttle)**.
6. In the Trackside tool, open the **Rock** page and pick a **Rock Switch** (section 7).



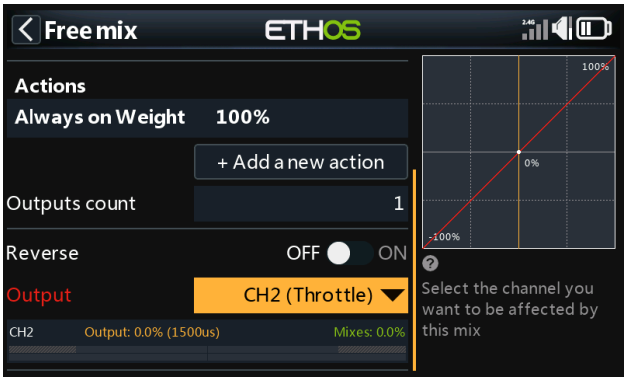
Model > Lua: Rock ON



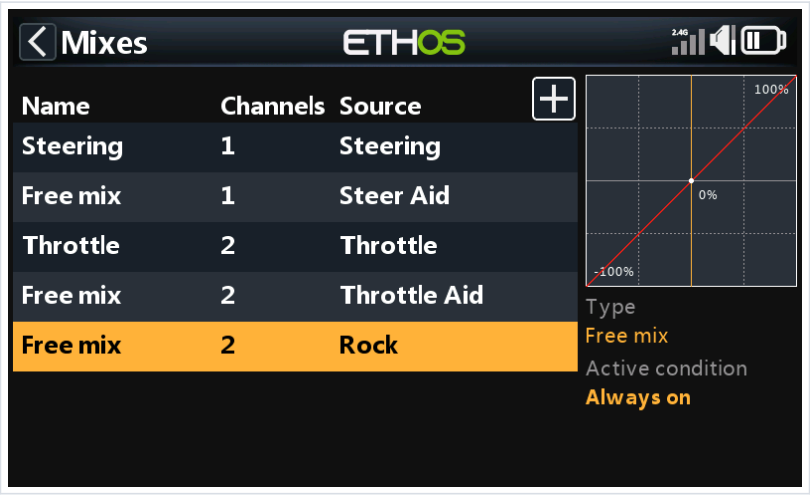
Source: Lua > Rock



Rock line: Operation Add, weight 100%



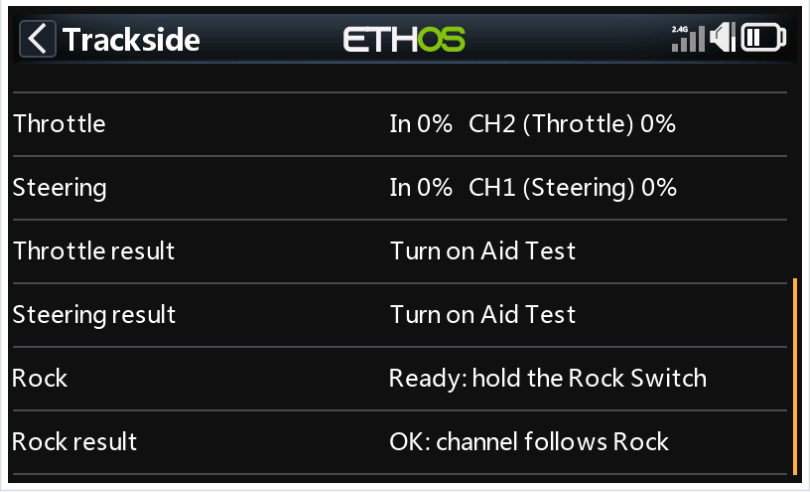
Output: CH2 (Throttle)



Finished: the Rock line is the last line on CH2

> The Rock line needs **no** safety switch. It adds nothing unless you are holding the Rock Switch, and if the Trackside scripts are missing it adds zero. It must be **Add**: with Multiply the throttle would always be zero.

Rock result on **Check setup** (section 5, at the bottom) needs no Aid Test. Hold the Rock Switch for a second or two with the trigger at rest, then let go. It should read **OK: channel follows Rock**. **No output** means the Rock line is missing, on the wrong channel, or not Add.

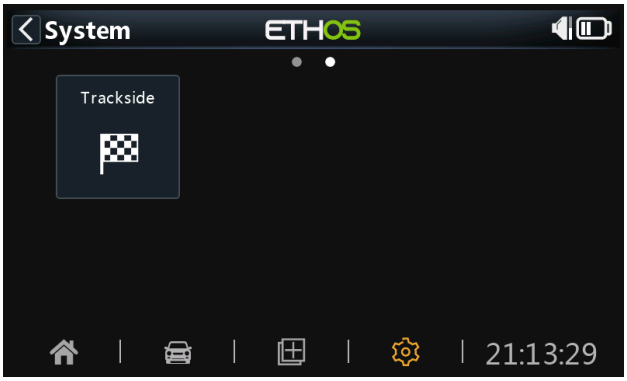


Rock result after holding the Rock Switch

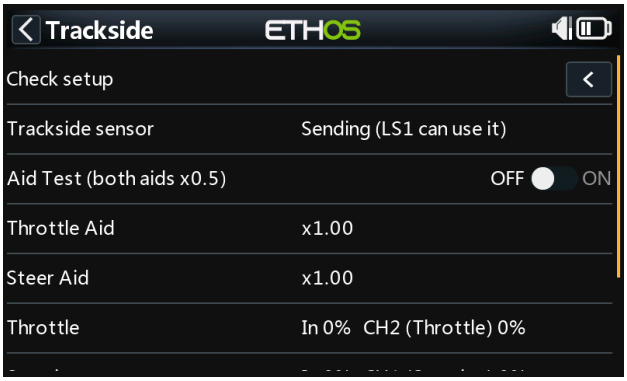
5. Check it works

With every aid OFF, a working setup looks the same as no setup. The **Aid Test** in the Trackside tool proves the mixes work: it pulses both aids between x1.0 and x0.5, so CH1 and CH2 swing between full and half travel. You can watch it on the Channels page too.

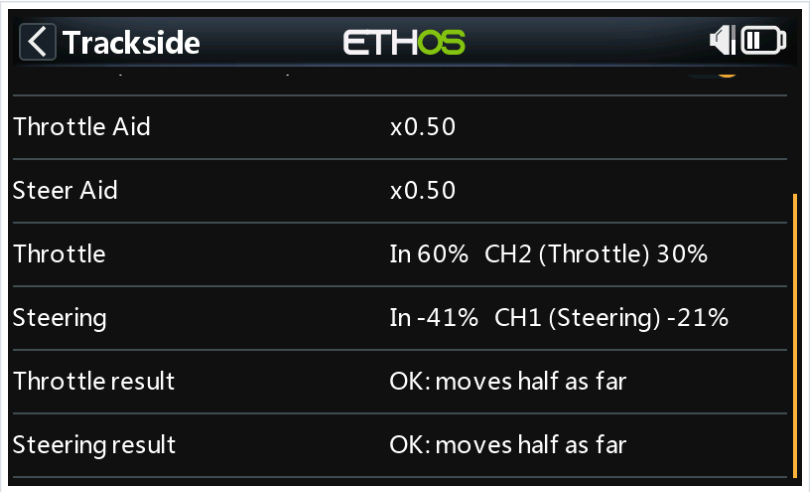
1. Open **System** (the gear in the bottom bar), swipe to the second page and tap **Trackside**.
2. Tap **Check setup**. **Trackside sensor** should read **Sending**, and both aids **x1.00**.
3. Turn **Aid Test** ON, then hold the trigger and the wheel steady past 30%.
4. Scroll down. Both results should read **OK: moves half as far**.
5. Leave the page with the back button. The Aid Test switches off by itself (also after 60 seconds).



System, page 2: the Trackside tool



Check setup, Aid Test OFF



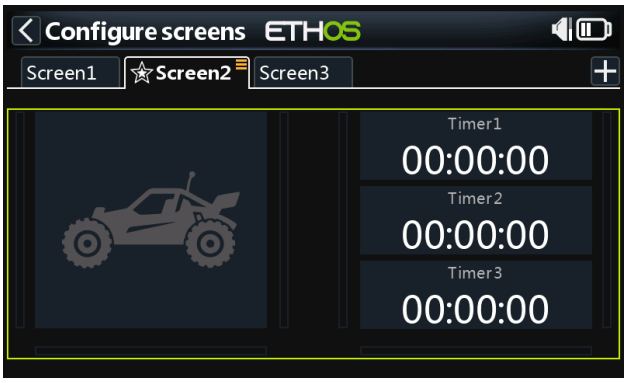
Aid Test ON: trigger 60% gives CH2 30%, both results OK

Result	Meaning and fix
OK: moves half as far	The aid line works.
Not working: aid line missing or off	The channel ignored the aid. Check the aid line exists, its Output is the right channel, it sits below the stick mix, and LS1 is green (section 4c).
Wrong: set Operation Multiply	The aid line is there but changes the channel the wrong way, usually Operation **Add**. Set it to **Multiply**.
Throttle Aid: Not running	Turn the aid on in **Model > Lua** (section 3).

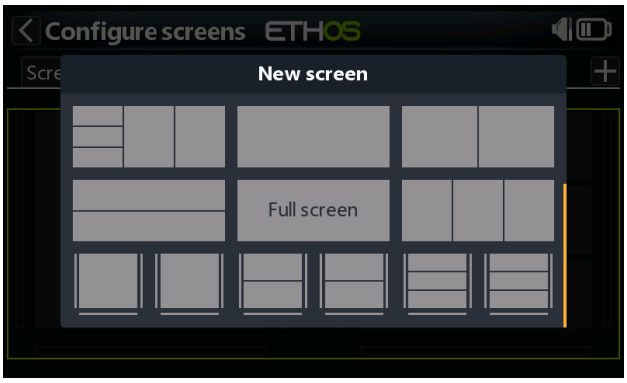
The check compares the channel to the input. If your aids drive other channels than CH1 and CH2, set **Throttle Channel** and **Steering Channel** on the tool's Inputs page.

6. Add the dashboard

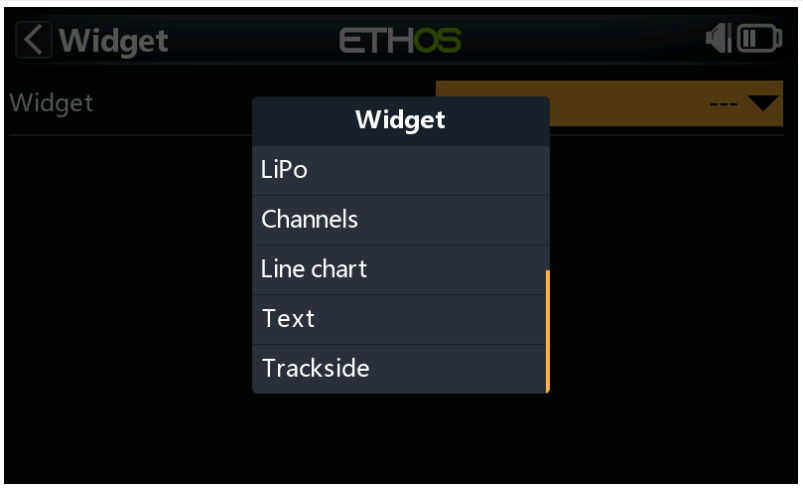
1. Tap the **Screens** icon in the bottom bar (the square with a +).
2. Tap **+** at the top right to add a screen.
3. Choose **Full screen**. Trackside also works in other layouts; in full screen it adds its own header with model name, link quality, radio battery and transmit power.
4. Tap the empty widget area, then pick **Trackside** from the widget list.
5. Go back to the home screen and swipe to the new screen.



Screens: tap + to add a screen



Choose Full screen



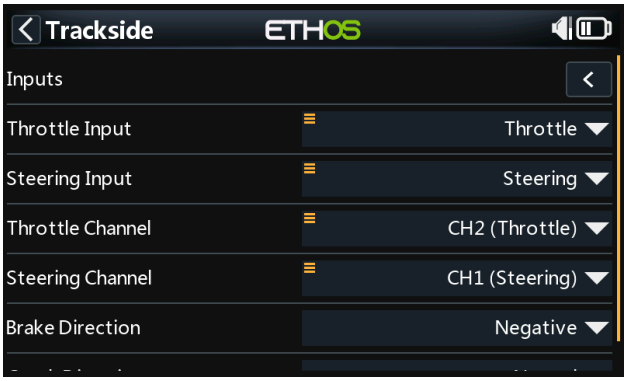
Pick Trackside from the widget list

7. Trackside settings

All settings are in the Trackside tool: **System**, second page, **Trackside**. Tap a tile to open its page; the back button returns to the menu. Settings are saved for the current model (its name is shown at the top) when you leave a page.



The Trackside tool: settings and diagnostics

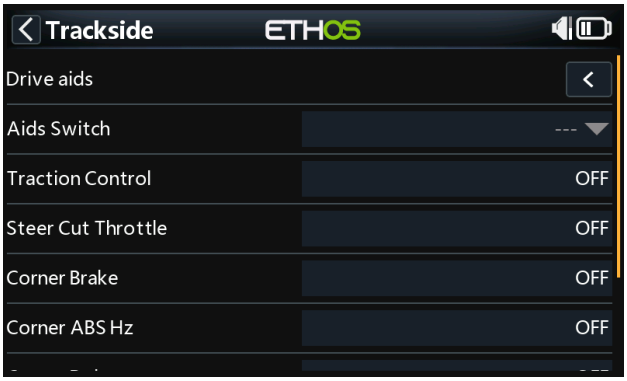


Inputs

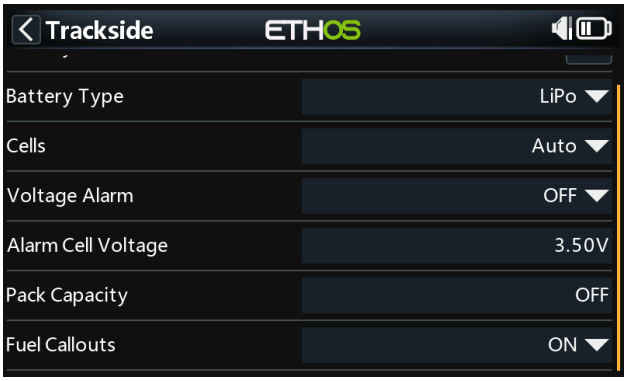
Check these first

Setting	Page	What to set
Throttle Input	Inputs	The control the aids read for throttle. Default: the VX10 trigger (Throttle).
Steering Input	Inputs	The control the aids read for steering. Default: the VX10 wheel (Steering).
Brake Direction	Inputs	Which way the trigger brakes. Pull the brake and watch the dashboard bar: it should turn red.
Battery Type, Cells	Battery	Match your packs (LiPo or LiHV). Cells = Auto detects the cell count.
Pack Capacity	Battery	Your pack size in mAh. Only needed if the car sends mAh used: fuel is then counted by mAh instead of estimated from voltage.
Fuel Callouts	Battery	Says "Fuel 50 percent" and so on at 50, 30, 20 and 10%. ON by default.
Voltage Alarm, Alarm Cell Voltage	Battery	Turn the low-voltage alarm on and choose the per-cell warning level.

Aids Switch (top of the Drive aids page): pick a switch position, for example **SA↑**. While it's in that position the aids work; in any other position every aid lets the sticks straight through. Leave it at **---** to have the aids always on. The LS1 safety switch from section 4c is still needed: it covers the case where Trackside isn't running at all.



Drive aids: Aids Switch at the top (aids OFF by default)



Battery: alarm, Pack Capacity, Fuel Callouts

The drive aids

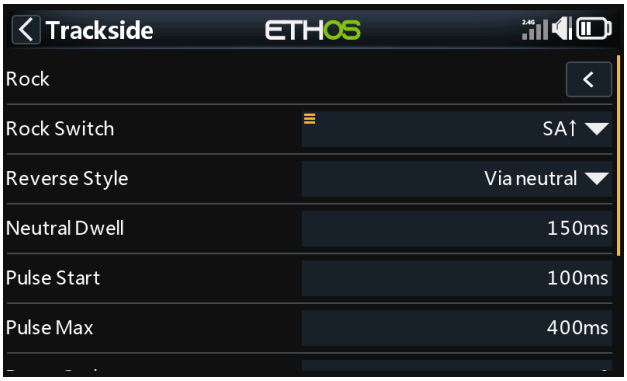
Aid (letter)	What it does	Try first
ABS (A)	Past ABS Trigger the brake pulses between full brake and Brake Floor. Dynamic pulses faster and lighter the harder you brake.	Static, 8 Hz, Brake Floor 40%, Trigger 50%
Traction Control (T)	Throttle can only rise so fast; letting off is instant. 0% = 0.1 s to full, 100% = 1 s.	30% on a slippery track
Steer Cut Throttle (S)	Less throttle the more you steer (100% = 60% less at full lock).	30%
Corner Brake (C)	Less brake the more you steer (100% = 60% less at full lock).	30%
Steer Limit (D)	Less steering at high throttle, from Steer Limit Start up (100% = 60% less at full throttle).	30% from 60%

Each letter on the dashboard lights in its own colour while that aid is changing your input. All aids are OFF by default. ABS above about 10 Hz can't be timed exactly by the radio.

Rock

Rock is OFF until you pick a **Rock Switch**. Use a spring-loaded switch or button if you have one. Hold it with the trigger at rest and the car rocks; let go and it stops at once. Pulling the trigger past 30% also stops it, and so does the **Hold Limit** ("Rock limit"). After it stops by itself, let go of the switch before trying again. If the switch is already in the Rock position when the radio starts, flick it off and back first.

Setting	What it does	Default
Rock Switch	Hold this switch position to rock. **---** = Rock off.	---
Reverse Style	Match your ESC. **Direct**: goes straight from forward to reverse (crawler ESCs). **Via neutral**: needs a pause at neutral before reverse (most ESCs). **Double tap**: needs brake, neutral, then reverse.	Via neutral
Neutral Dwell	How long each pause at neutral lasts. Raise it if the car only brakes.	150 ms
Pulse Start, Pulse Max	Length of each push, from the first rock to the longest.	100, 400 ms
Power Start, Power Max	Throttle of each push, from the first rock to the strongest.	60, 100%
Ramp Cycles	Rocks it takes to go from Start to Max. Then it starts again from Start.	6
Hold Limit	Rock stops by itself after this long, to protect the motor and ESC.	6 s



The Rock page



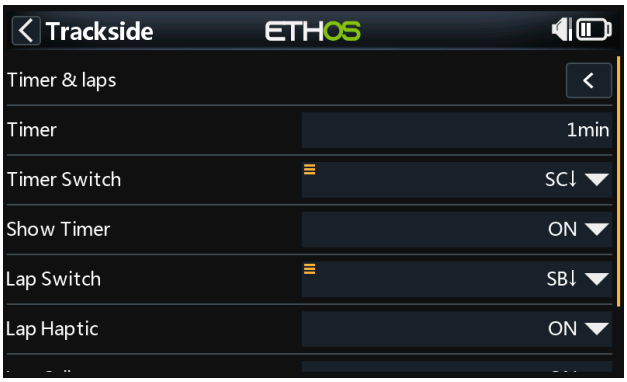
Rocking: R lit, the bar shows each push

Start with the defaults. If the car only brakes, set Reverse Style to **Double tap** or raise Neutral Dwell. If it rocks but never goes over, raise Power Max or Pulse Max. If it flips too hard, lower Power Start.

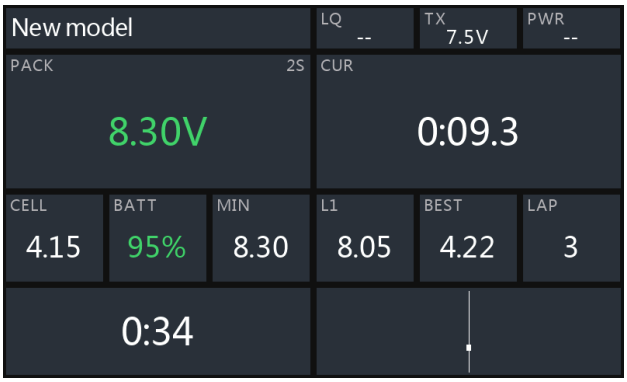
Laps and timer

On the **Timer & laps** page, set **Lap Switch** to a button or switch position you can press as you cross the line, and **Timer Switch** to another. Turn **Show Timer** ON to see the countdown in place of the aid letters.

Do this	What happens
Press the Lap Switch	The first press starts the clock (or the first throttle, with Throttle Start Lap ON). Each later press ends a lap: buzz, and the lap time is spoken ("Best lap" first on a new best). Presses less than 3 seconds apart are ignored.
Hold the Lap Switch 1.2 s	A bar fills along the CUR box, then the laps reset ("Laps reset").
Press the Timer Switch	Starts or pauses the countdown. After it has run out, starts it again.
Hold the Timer Switch 1.2 s	Resets the countdown to Timer minutes ("Timer reset").

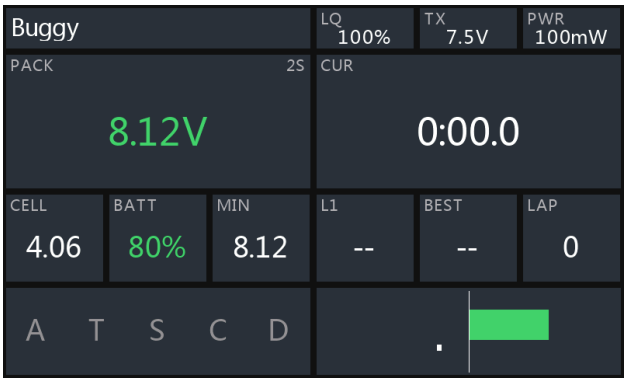


Timer & laps settings

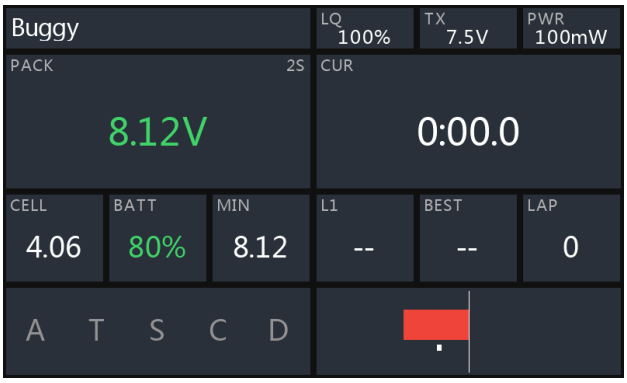


Racing: lap 3 running, best 4.22, countdown in the footer

8. Reading the dashboard



On throttle: the bar fills green to the right



On the brake: the bar fills red

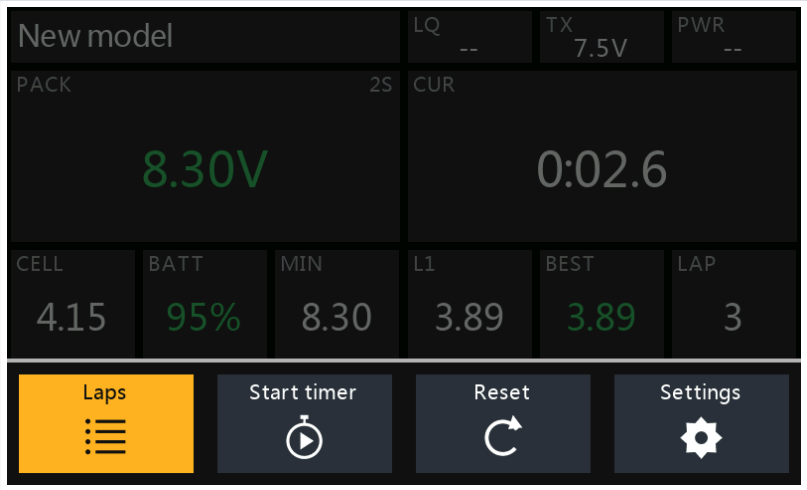
Area	Shows
Header (full screen)	Model name, link quality (LQ), radio battery (TX) and transmit power (PWR).
PACK	Car pack voltage, with the detected cell count (for example 2S) in the corner. Green above 50% fuel, amber below 50%, red below 20%. With Voltage Alarm ON: amber while the cells are under the alarm level, and the box flashes red once the alarm goes off.
CELL / BATT / MIN	Voltage per cell, fuel left (see below), and the lowest pack voltage this session.
CUR / L1 / BEST / LAP	Lap timing: current lap, last lap, best lap (green for 3 s after a new best) and lap count.
A T S C D R	Drive aid letters: in the aid's own colour while it's working, white when it's armed, grey when it's OFF. Struck through when the Aids Switch has turned the aids off. **R** (Rock) lights pink while rocking, whatever the Aids Switch. With Show Timer ON this box shows the countdown instead: grey when paused, red and blinking when it has run out.
Live bar	Throttle (green) or brake (red) from the centre line. The small mark below shows the wheel position. While rocking it shows each push: green forward, red reverse.

Telemetry values show **--** until the receiver sends them.

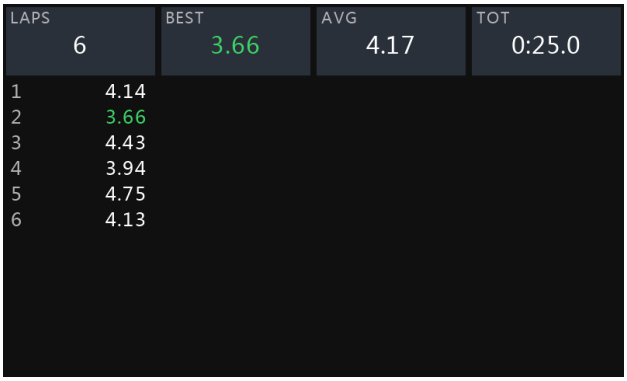
The toolbar

Swipe up on the dashboard (or long-press PAGE) to bring up the toolbar. Choosing a button closes it again; so do a swipe down, a tap above it, RTN, or 10 seconds without a touch. Swiping sideways between screens works as usual.

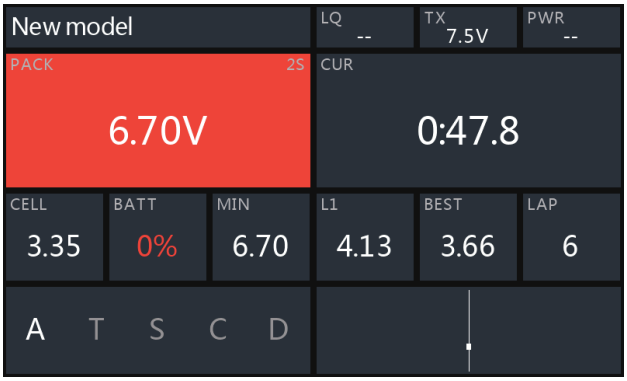
Button	Does
Laps	Opens the lap view: every lap of the session, best in green, with LAPS, BEST, AVG and TOT. Tap anywhere (or RTN) to go back.
Start / Pause timer	Starts or pauses the countdown (restarts it after it has run out).
Reset	Asks, then resets the laps and the countdown.
Settings	Opens the Trackside tool (needs Ethos 26.1 or later).



The toolbar, after a swipe up



Lap view: 6 laps, best 3.66



Battery alarm: the PACK box flashes red

Fuel (BATT)

Source	Used when	How it works
Fuel sensor	The car sends a fuel sensor (for example **Fuel** from the ESC)	Shown as sent. Trackside always prefers it.
mAh used	Pack Capacity is set and the car sends mAh used	Starts from the resting voltage, then counts down by mAh used.
Voltage	Otherwise	Read from the resting voltage whenever you've been off the throttle for a second. On throttle it holds, and once you've driven it never goes back up.

For the mAh and voltage sources Trackside creates a telemetry sensor called **Pack Fuel**, so you can use it in logic switches, alerts or other widgets. Check setup shows which source is in use.

Spoken alerts

You hear	When
Fuel 50 percent (30, 20, 10)	Fuel passes each level, once per pack. Fuel Callouts on the Battery page.
Battery low, 3.4 volts + vibration	Voltage Alarm is ON and the resting cell voltage stays under Alarm Cell Voltage for 5 seconds (readings on throttle don't count). Repeats every 30 seconds.
Aid test on / Aid test off	The Aid Test on Check setup starts or ends.
Drive aids on / Drive aids off + short buzz	You flip the Aids Switch.
4.2 seconds / Best lap, 4.2 seconds + buzz	Each lap (Lap Callout and Lap Haptic on the Timer & laps page).
One minute / Time up + vibration	The countdown reaches 1:00 and 0:00. After 0:00 it buzzes every 10 s for a minute.
Drive aids off. Trackside error. + vibration	A Trackside error released the aids: the car drives on the plain sticks.
Short buzz	Rock starts, or stops because you pulled the trigger.
Rock limit + vibration	Rock stopped by itself at the Hold Limit.

Alerts use the radio's voice setting (English: default, US or UK). To use your own recording, put a WAV with the same name in **SCRIPTS/trackside.user/audio**.

9. Troubleshooting

Problem	Check
No steering or throttle at all	Trackside isn't loaded and the aid lines have no safety switch. Set both aid lines' Active condition to **LS1** (section 4c), then install Trackside. Also check the Steering and Throttle mixes still use Analogs > Steering and Analogs > Throttle as their source.
Throttle or steering moves only a little	Update Trackside to the latest version. Early test builds scaled the aid output wrongly.
The aid mix has no effect	Run **Check setup** with the Aid Test (section 5): it says what is wrong. Set the Free mix's **Output** to the channel (CH2 or CH1), check that Operation is **Multiply**, and that the aid line sits **below** its stick mix. LS1 must be green (Model > Logic switches).
The aids stopped working after reinstalling Trackside	If the model was changed while Trackside was missing (even a trim click), Ethos clears its Lua links. The car still drives normally. Open each aid line and pick its Lua source again. LS1 and the Trackside sensor are kept.
No Lua category when picking a source	Turn on Throttle Aid and Steer Aid in **Model > Lua** (section 3). This is needed once per model.
Trackside isn't in the widget list	Check the folder is **SCRIPTS/trackside** with **main.lua** inside, then restart the radio.
The dashboard shows -- everywhere	No telemetry yet. Values appear once the receiver is bound and sending telemetry.
The bar turns red when I accelerate	Change **Brake Direction** on the tool's Inputs page.
Rock does nothing	Check the **Rock result** on Check setup (section 5). Rock needs a Rock Switch, the trigger at rest, and the switch to have been off once since the radio started.
Rock only pushes forward (the car brakes instead of reversing)	Turn reverse on in the ESC. Then try Reverse Style **Double tap**, or raise Neutral Dwell.